



Quantum enhanced stratification of breast cancer: exploring quantum expressivity for real omics data

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Abstract

Quantum machine learning (QML) is considered one of the most promising applications of quantum computing in the noisy intermediate-scale quantum (NISQ) era for the impact it is thought to have in the near future. Although promising theoretical assumptions, the exploration of how QML could foster new discoveries in medicine and biology fields is still in its infancy with few examples. In this study, we aim to assess whether quantum kernels (QK) could effectively classify subtypes of breast cancer (BC) patients based on molecular characteristics. We performed a heuristic exploration of encoding configurations with different entanglement levels to determine a trade-off between kernel expressivity and performances. Our results show that QKs yield comparable clustering results with classical methods while using fewer data points and are able to fit the data with a higher number of clusters. Additionally, we conducted the experiments on the quantum processing unit (QPU) to evaluate the effect of noise on the outcome. We found that less expressive encodings showed a higher resilience to noise, indicating that the computational pipeline can be reliably implemented on NISQ devices. Our findings suggest that QK methods show promises for application in precision oncology, especially in scenarios where the dataset is limited in size and a granular non-trivial stratification of complex molecular data cannot be achieved classically.

Keywords Quantum machine learning · Quantum kernel · Clustering · Breast cancer

1 Introduction

Quantum machine learning (QML) has been proposed as one of the most promising and worth investigating applications of quantum computing (QC) for current NISQ devices (Biamonte et al. 2016; Dunjko and Briegel 2018; Perdomo-Ortiz et al. 2017; Schuld 2021). In the QML framework, classical ML algorithms are embedded into the quantum-mechanical formalism by exploiting the paradigm shift of QC's information. The quantum processing leverages entanglement, superposition, and interference to enhance, speed up, and optimize over classical computing, sometimes even exponentially faster (Bernstein and Vazirani 1993; Shor 1994). This can help in learning in both computational and sample complexity aspects (Huang et al. 2020). Nevertheless, the quantum embedding alone does not transversally guarantee learnability (Park et al. 2020; Gil-Fuster et al. 2023; Slatery et al. 2023) in particular for real classical data (Cerezo et al. 2022). Classical ML methods thrive to find and learn patterns in the data, and they were successfully applied to

different fields, including biology and medicine. Specifically for precision oncology, ML methods are crucial in specific tasks, such as prognostic predictions, human cancer stratification, and discovery of new clinical subgroups using omics datasets (Tran et al. 2021). Nevertheless, despite the substantial progress facilitated by computational methods, some fundamental constraints in modeling biological and clinical systems still persist. In fact, ML models often fail to capture and reconcile the real and complex dynamics of tumors due to reductionist approaches to the problem. Moreover, due to the high heterogeneity of complex omics information, available data can not be enough for classical ML models to properly solve their structure at a granular level (Cordier et al. 2021). Facilitated by the technological innovation and rapid reduction in sequencing costs, thousands of cancer samples have been profiled so far by pan-cancer studies (Weinstein et al. 2013) revealing high level of heterogeneity both intra- and inter-samples. The inter-sample heterogeneity influences clinical tumor classification, patient prognosis, and prediction of response to personalized strategies (Caswell-Jin et al. 2021). Hence, it is of pivotal importance to take it into account, for an effective

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tive application of precision oncology. Breast cancer is of special interest given its high prevalence (Siegel et al. 2020) and its complex landscape of somatic alterations which gave rise to multiple and divergent subtype classifications (Curtis et al. 2012; Russnes et al. 2017). In this respect, it represents an ideal complex and real use case to investigate the potentiality of QML approaches to disentangle molecular heterogeneity and extract meaningful non-trivial patterns and relationships within the considered dataset (Flother 2023). Indeed, the exploration of QML approaches in biology and medicine is still at an early stage, with most proof of concept studies exploiting QML models for downsized problem in diagnostics and treatment (Flother 2023). In particular, quantum support vector machine (QSVM) has been employed as supervised classification method mainly for distinguishing pathological conditions using clinical records (Krunić et al. 2021) or biomedical images (Ahalya et al. 2022). For cancer case, BC was classified from mammography using quantum transfer learning (Azevedo et al. 2022), from ultrasound images with quantum neural network (Amin et al. 2022), and from digitized image features of fine needle aspirate (Moradi et al. 2021; Vashisth et al. 2021) or digital breast tomosynthesis (Aswiga et al. 2024) using QSVM. Very few attempts have been made so far with large and diverse datasets as those including omics data. To the best of our knowledge, the only study is by Li et al. (2019) who trained a QSVM exploiting annealing based Ising ML algorithms on human tumor data from the TCGA and compared its performance with conventional ML algorithms in distinguishing cancer versus normal healthy samples or, among BC tumors, in stratifying by hormonal levels (i.e., LumA vs LumB, ERpos vs ERneg), and in reducing sample complexity. Among the possible quantum algorithm that promise to be disruptive for treating complex data, quantum kernels (QKs) are regarded as highly efficient tools for identifying non-linear data relationships, by measuring the similarity in a given higher dimensional representation space, which is native when encoding input data in a quantum state (Schuld and Killoran 2018). In this study, we make use of QK methods with complex multi-omics tumor data, aiming to test their potential to detect a more granular structure of the tumor samples and seek new subgroups, approaching the problem in an unsupervised way. Since the embedding of classical data in a proper way is a highly non-trivial task (Incudini et al. 2022), we investigated different QK set-ups that modulate the expressivity of the embedding itself. We also tested various sampling of the BC multi-omics dataset, running the experiments on both simulated and real quantum hardware to assess the consistency of our findings, and finally, we compared them with the classical kernel counterpart. The results show that the quantum approach foster a clusterization that better fit the data with higher number of cluster. Moreover, we noticed that performances trend remains consistent across sample sizes with

higher coherence for higher scoring quantum embeddings, suggesting that, for those cases, the patterns extracted can be learned with fewer data points. We furthermore demonstrated that quantum noise, which in principle hinders the quantum computation, has little or no effect on the computational pipeline for some configurations, thus showing that, in principle, the approach shown here can be implemented on NISQ devices for practical real-world applications.

2 Background

2.1 Kernel theory

In kernel theory, given a set of training inputs x belonging to the input space $X \in \mathbb{R}^n$, to recognize patterns, a similarity measure called kernel k (Williams 2001) is defined as a function

$$k : X \times X \rightarrow \mathbb{R} \quad (1)$$

Let $\mathcal{H}$ be a Hilbert space, called the feature space, and x a sample of an input set X . We can define a feature map Φ

$$\Phi : X \rightarrow \mathcal{H} \quad (2)$$

as a map from inputs to vectors in the Hilbert space. The inner product of two inputs embedded in the feature space define a kernel k :

$$k(x_i, x_j) := \langle \Phi(x_i), \Phi(x_j) \rangle \quad (3)$$

For a set of N input points $\in X$, a positively definite matrix called kernel matrix (or Gram matrix) $K \in \mathbb{R}^{N \times N}$ is defined using the kernel function k

$$K := [k(x_i, x_j)]_{ij} \quad (4)$$

for all $(x_i, x_j) \in X$. The power of kernel methods is that the feature space is usually of much higher dimension than the original space, sometimes even of infinite dimension, where however the inner product can still be computed through efficient manipulation of input data. An effective and commonly use example is the radial basis function (RBF) kernel that projects the input in an infinite dimensional feature space by evaluating the kernel matrix as follows:

$$K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|_2} \quad (5)$$

where $\|x_i - x_j\|_2$ is the Euclidean distance between the two input vectors. When the parameter $\gamma = \frac{1}{\sigma^2}$, the kernel is known as the Gaussian kernel, with σ being the variance of the data. QKs are kernels where the kernel is evaluated

through inner product of quantum states, since the process of encoding inputs in a quantum state is interpreted as a feature map that embed data to the quantum Hilbert space (Schuld and Killoran 2018). In QK, the input data $x \in R^n$ are encoded, via an encoding function $\phi(x)$, in a quantum state. Formally, this encoding process is realized through the application of a unitary transformation $U_\phi(x)$ acting on n qubits in the initial ground state $|0\rangle^{\otimes n}$ (Mengoni and Pierro 2019):

$$\phi : \mathbf{x} \longrightarrow |\phi(x)\rangle = U_\phi(x)|0\rangle^{\otimes n} \quad (6)$$

The kernel matrix is then computed as the pairwise inner product of each quantum state in the dataset:

$$\begin{aligned} K(x_i, x_j) &= |\langle \phi(x_i) | \phi(x_j) \rangle|^2 \\ &= \langle 0 |^{\otimes n} U_{\phi(x_i)}^\dagger U_{\phi(x_j)} | 0 \rangle^{\otimes n} \end{aligned} \quad (7)$$

where the QK matrix entries are highly influenced by the choice of the unitary transformation U_ϕ (Suzuki et al. 2019).

2.2 Quantum embedding of classical data

Classical data can be encoded into the state $|\phi(x)\rangle$ in several ways and the design of $U_\phi(x)$ should be chosen to maximally exploit the quantum setting, yielding to classically intractable kernels that are hard to simulate but easy to implement on a quantum computer (Incudini et al. 2024). For this reasons, we adopted same IPQ-style encoding as in (Havlicek et al. 2018). More formally, the unitary ansatz we employed is generally described as follows:

$$U_\phi(x) = U_Z(x)H^{\otimes n} \quad (8)$$

where there is a layer of Hadamard gates H acting in parallel on the n qubits, followed by a unitary U_Z that, for entangled embeddings in Havlicek et al. (2018), takes the form:

$$U_Z(x) = \prod_{[i,j] \in S} R_{Z_i Z_j}(x_i x_j) \bigotimes_{k=1}^n R_Z(x_k) \quad (9)$$

where R_{ZZ} is a two qubits rotation that makes them entangled and S is the set of qubits to be entangled. Nevertheless, this choice does not guarantee alone an advantage over classical embedding as it is deeply influenced by the data characteristics and tasks (Incudini et al. 2022). In this respect, we employed the geometric difference ($g_{C|Q}$), introduced by Huang et al. (2020), to evaluate if the classical and quantum kernels implemented differ. This is an anti-symmetric difference between quantum and classical kernel functions that allows to perform a prescreening, independent of the labels, to measure the separation between the two models

for the particular dataset used. The specifics of the quantum embeddings chosen for our experiments are described in the next “Experimental set-up” section.

2.3 Expressivity regularization

The inherent high dimensionality of the Hilbert space used for projecting the input data can hinder the model performance. In particular for QK, it has been shown that, as a result of highly expressive problem-agnostic encoding, the non-diagonal entries distribution often skews around the mean (Thanasilp et al. 2022), making the points in the quantum Hilbert space nearly indistinguishable. This phenomenon has a close parallelism to the curse of dimensionality of classical ML. It has been shown (Shaydulin and Wild 2022; Canatar et al. 2022) that acting on the parametric rotational gate angles can regularize the model expressivity. This strategy is implemented with a tunable hyperparameter called *bandwidth*, which is a constant within the range $[0, 1]$. This parameter effectively scales the input, confining the embeddings to a more restricted region of the quantum state space as in:

$$|\phi(x)\rangle = U_\phi(cx)|0\rangle^{\otimes n} \quad (10)$$

where c is the bandwidth parameter.

3 Experimental set-up

3.1 Breast cancer multi-omics dataset

The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) dataset (Curtis et al. 2012) includes the multi-omics profiles of 1980 primary biopsy samples of BC patients and represents the first large effort for studying the BC molecular heterogeneity. METABRIC dataset offers valuable insights, as it comprehensively encompasses a variety of multi-omics profiles which include gene expression and copy number variants. These two omics count respectively 20,603 and 22,544 features and have highly distinctive properties: gene expression levels are positive continuous variables indicating the abundance of mRNA molecules, whereas CNVs are categorical features representing somatic alterations which involve the changes in the number of copies of a particular DNA segment. In addition, intra-modal and inter-modal independence is not satisfied, and both modalities are subject to the intrinsic noise of the experimental procedure employed to extract these molecular features (Bersanelli et al. 2016).

The integration of the genomic and transcriptomic profiles revealed 10 subtypes, termed integrative clusters (IntClust), characterized by distinct genomic drivers (Curtis et al. 2012).

These subtypes hold significant clinical relevance, as they exhibit varying patterns of survival (Curtis et al. 2012). It is important to note that multiple interpretations of the same dataset have been proposed so far based on alternative intrinsic subtypes (e.g., luminal A, luminal B, HER2-enriched, basal-like or triple-negative, and normal-like), and therefore, the IntClust categorization should not be viewed as overly restrictive (Caswell-Jin et al. 2021).

Due to the high dimensionality of the dataset, we performed a reduction step formally projecting our input data in a lower dimensional space (i.e., 4-dimensional). To avoid flattening of meaningful signals through this stage, we used Uniform Manifold Approximation and Projection (UMAP) (McInnes et al. 2018) as a reduction strategy since it exploits topological data analysis (TDA) ensuring the topology of the underlying data is captured and maintained in the reduced space. The algorithm effectively uses a set of hyperparameters to balanced local and global structure; for our case we implemented the transformation with the default parameters of the UMAP library.

3.2 Data encoding: quantum and classical

The choice of a suitable quantum embedding, in case of complex real data, is not a straightforward task (Cerezo et al. 2022). Nevertheless, it is possible to explore, as set out in this work, a subset of configurations of the encoding scheme, in particular by varying the entanglement map of the unitary ansatz. In this way, a trade-off between the expressivity of the kernel and the task-related performance can be assessed. We specifically explored three configurations of the IPQ-style encoding scheme, making use of unitary templates with different levels of entanglement of the qubits. The most simple configuration employed is the Z feature map without any

entanglement, utilizing a sequence of Hadamard and rotational P gates, a single qubit rotation around the Z axis, implemented in our case for a 4-dimensional classical input x as shown in Fig. 1a. We then implemented the ZZ feature map with both linear and full entanglement: In the linear entanglement, each qubit is entangled with its neighboring qubits, while in the full entanglement, each qubit is entangled with every other qubit. These feature maps resulted respectively in the embedding shown in Fig. 1b and c. Furthermore, we regularized the expressivity of the quantum embeddings tuning the angle range β . For each feature map, we tested five different scaling of the angle range so that the input data $x \in [0, \beta]$, where $\beta \in \{\frac{\pi}{8}, \frac{\pi}{4}, \frac{\pi}{2}, \pi, 2\pi\}$. All the feature maps have been implemented using Qiskit (Aleksandrowicz et al. 2019).

To compare the quantum embeddings with a classical kernel, we compute the RBF kernel, expressed by Eq. 5, using the sklearn implementation with default parameter $\gamma = \frac{1}{n\sigma}$ where n is the number of feature (Pedregosa et al. 2011). We selected the RBF kernel due to its superior expressiveness compared to other classical kernels, enabling it to effectively capture complex, non-linear relationships in the data (Scholkopf and Smola 2001).

3.3 Clustering evaluation

Clustering is a widely used approach in complex biological data analysis to group similar samples based on molecular descriptors and to uncover relationships between cases. Here, we choose to adopt spectral clustering method to stratify BC samples as it relies on the similarity matrix of the input data like the one we obtain from the kernel matrix (Eq. 7). Diverse internal validation metrics exist to evaluate clustering robustness in an unsupervised case. Each of them is differently

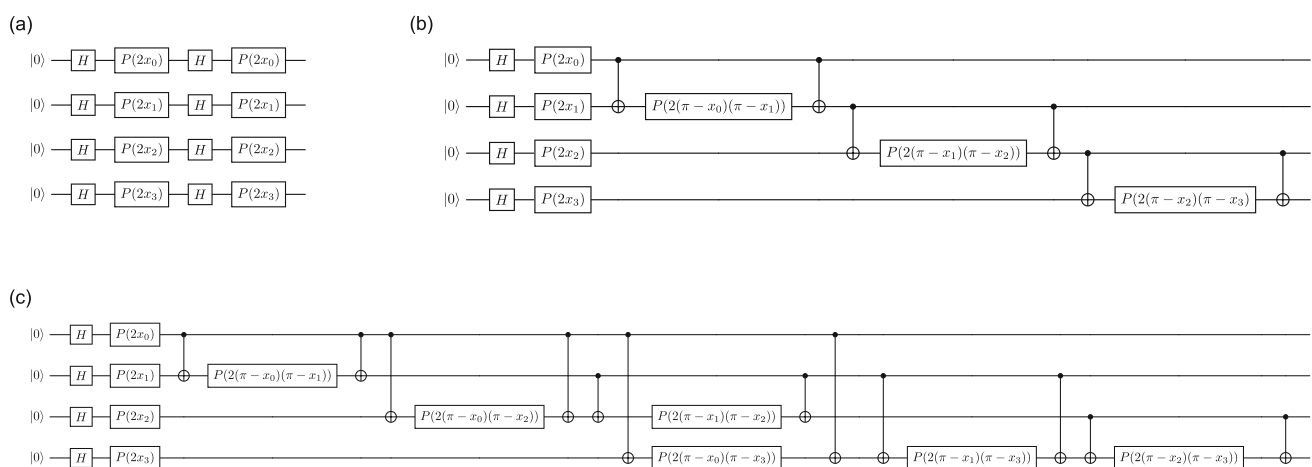


Fig. 1 Graphic high level circuit representation of the four qubits feature maps used with different level of entanglement: **a** Z feature map with no entanglement, **b** ZZ feature map with linear, and **c** full entanglement

influenced by data characteristics (i.e., noise, density, sub-clusters, skewed distributions) (Liu et al. 2010) and has to be interpreted by considering which entities are depicted by the data, as for biological dataset (Ronan et al. 2016). In our case, with access only to the kernel matrix, we opted for the silhouette coefficient (SC) (Rousseeuw 1987) which relies solely on the pairwise distance between points and is defined as follows:

$$SC = \frac{1}{N} \sum_i s(i) \quad (11)$$

given a data set of N points and $s(i)$ being the silhouette value associated with the i -th point which is defined as follows:

$$s(i) = \frac{b(i) - a(i)}{\max_i(a(i), b(i))} \quad (12)$$

in which $a(i)$ and $b(i)$ are respectively the average distance of the i -th point to its own cluster and the average distance to its second closest cluster. The value of the both the SC and the $s(i)$ ranges in the interval $\{-1, 1\}$ where a high value indicates that the object is well matched to its own cluster and weakly matched to neighboring clusters. In addition, we employed adjusted mutual information (AMI) as a measure of agreement between two sets of labeled data (Nguyen et al. 2009). Given two partitions U and V of the same set of N points, the AMI between them is calculated as follows:

$$AMI(U, V) = \frac{MI(U, V) - E\{MI(U, V)\}}{\max\{H(U), H(V)\} - E\{MI(U, V)\}} \quad (13)$$

where H is the Shannon entropy over R clusters and MI is the mutual information between the two partitions, respectively, defined as follows:

$$H(U) = - \sum_{i=1}^R P_U(i) \log P_U(i) \quad (14)$$

$$MI(U, V) = \sum_{i=1}^{|U|} \sum_{j=1}^{|V|} \frac{|U_i \cap V_j|}{N} \log \frac{N|U_i \cap V_j|}{|U_i||V_j|} \quad (15)$$

4 Results

4.1 Exact simulation

We analyzed the performance of the chosen quantum embeddings in an ideal noiseless setting on the full data set with 1980 samples and compared the results with a classical RBF embedding. In this phase, we computed the QKs on a locally simulated quantum computer with 4 qubits, using the *qasm*

simulator backend from Qiskit, which is designed to simulate the behavior of an ideal quantum circuit. For each feature map, the QKs have been evaluated for input values within the range $[0, \beta]$, with $\beta \in \{\frac{\pi}{8}, \frac{\pi}{4}, \frac{\pi}{2}, \pi, 2\pi\}$. We investigated the difference between the RBF and each quantum kernel function by computing the $g_{C|Q}$, using the implementation provided by Di Marcantonio et al. (2023). The $g_{C|Q}$ is used as a prescreening of quantum embedding, in particular a large $g_{C|Q}$ compared to the threshold $\sqrt{N}$ indicates a significant deviation between the performances of the two kernels (Huang et al. 2020). The results in Fig. 2 demonstrate that the selected quantum embeddings exceed this threshold, highlighting a significant deviation from the classical counterpart. This behavior correlates with the expressivity of the encoding: the entangled ZZ linear and ZZ full feature maps show $\beta \geq \frac{\pi}{4}$ have a large $g_{C|Q}$ whereas the unentangled Z feature map exceeds the threshold only for less regularized embedding with $\beta \geq \pi$. Considering no assumptions were made regarding the partition of our data, for the clustering, we optimized the number of clusters (K) in a range from 2 to 10, where $K = 10$ refer to the IntClust subtypes (Curtis et al. 2012). Specifically, we measured the SC, as defined in Eq. 11, for each feature map (i.e., quantum and RBF). The results of the SC for the exact simulations are shown in Fig. 3. Overall, it can be noticed that ZZ linear map outperforms the classical RBF model for $K = 2$, the trivial case, and for finer grouping with $K \geq 9$, with $\beta = \frac{\pi}{8}$ and with $\beta = \frac{\pi}{4}$, respectively. However, the RBF kernel shows higher SCs for $3 \leq K \leq 8$. This seemingly irregular behavior can be explained taking into consideration that the SC might be affected by the presence of macro and micro structures within the data, advantaging clustering with lower K (Liu et al. 2010). The SC of ZZ full also shows an improvement as K increases for all β values, although significantly lower than ZZ linear case, suggesting that a more granular stratification better suits the data structure. The Z feature map, on the other hand, favors clustering

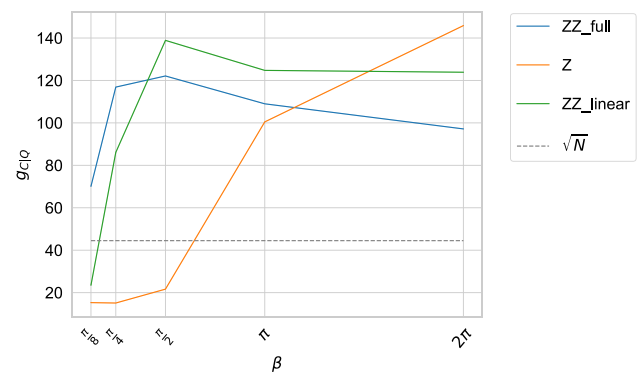


Fig. 2 Geometric difference $g_{C|Q}$ between classical and quantum kernels of the full dataset ($N=1980$) plotted against the angle range β and colored by feature map. The dashed gray line denotes the threshold for a significant difference between the kernels

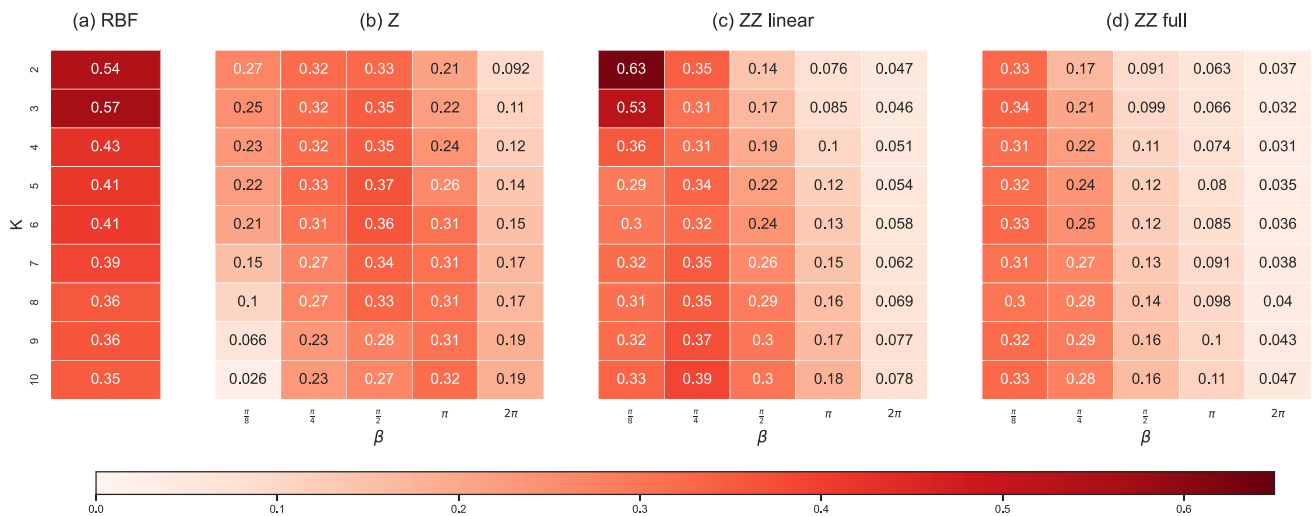


Fig. 3 Heatmaps with the SC at varying numbers of clusters K for each feature map configuration: **a** classical RBF; **b** Z encoding; **c** ZZ with linear entanglement; **d** ZZ with full entanglement. For quantum feature maps, SC values are reported at varying of the maximum angle range β

with lower K for $\beta \leq \frac{\pi}{2}$ being $K = 5$ the best SC value. The trend then switches for $\beta > \frac{\pi}{2}$ for which higher values of K yield a better SC. In addition, the experimental results demonstrate that the quality of the clustering, entailed by the SC, depends on the complexity of the encoding, i.e., on the entanglement structure of the feature map, and further by the angle β , which provides a regularization of the expressivity. In fact, the entangled ZZ feature maps strongly benefit from regularization, as they show a subsequent decrease of the score with the increase of the angle range β , whereas the Z feature map provides its best results with some level of angle tuning for $\frac{\pi}{4} \leq \beta \leq \pi$ and has its pick for $\beta = \frac{\pi}{2}$.

4.1.1 Sample complexity analysis

To further assess the sample complexity of quantum models for our data, we focused on the best-performing ZZ linear feature map. We selected the highest scoring embedding configurations by setting a threshold of $SC \geq 0.3$ and inspected how samples are grouped among clusters when the sample size varies from 1980, the full set, to a subset of N samples with $N \in \{50, 100, 250, 500, 800, 1000\}$. We estimated clustering coherence of the reduced sets by calculating the AMI with respect to the full data for both classical RBF and quantum feature maps. The results are reported in Fig. 4. On

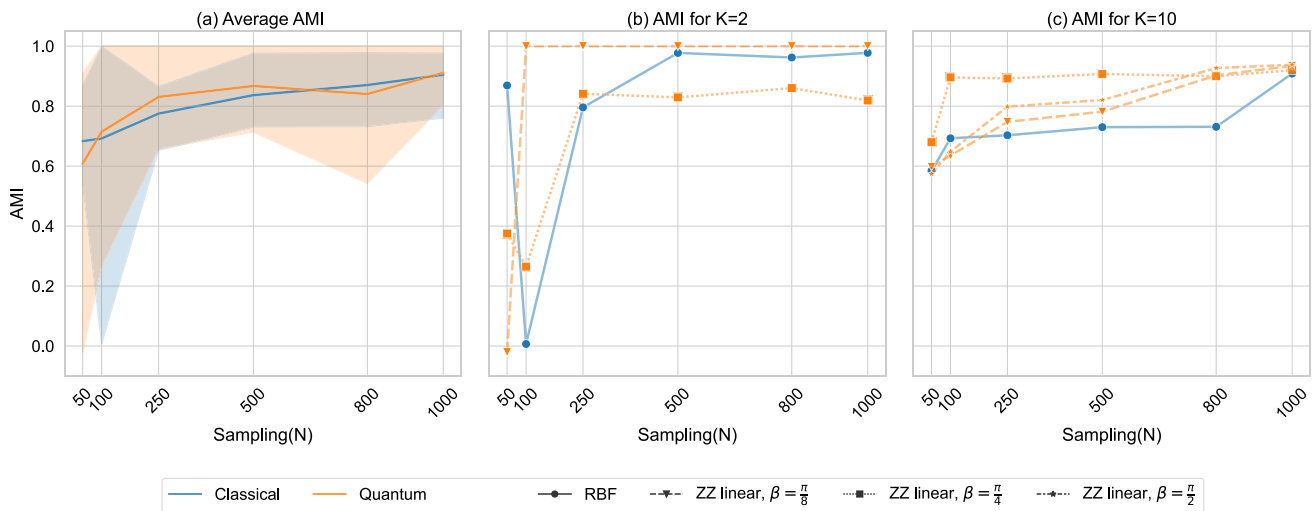


Fig. 4 Adjusted mutual information (AMI) scores as a function of data sampling. Results compare classical (blue) and quantum (orange) stratifications for configurations with $SC \geq 0.3$. **a** Mean AMI scores aggregated by kernel type (classical and quantum), with error bars rep-

resenting the range (maximum and minimum AMI) for each kernel type. **b** AMI scores for configurations with $K = 2$. **c** AMI scores for configurations with $K = 10$

average, the quantum-derived clustering reaches a plateau with an $AMI \geq 0.8$ at $N = 250$ (Fig. 4a), slightly before the classical one. For the trivial case $K = 2$, which presents the highest SCs for both the classical and the quantum case, the ZZ linear map with $\beta = \frac{\pi}{8}$ reaches perfect agreement ($AMI = 1$) at $N = 100$ whereas the RBF kernel needs $N = 500$ samples (Fig. 4b). Similar outcomes can be observed with a more granular grouping of the samples with still high SC (i.e., $K = 10$), in which the best configurations of quantum clustering achieve higher AMI compared to the classical one for each sample subset with $N < 1000$. Moreover, the ZZ linear with $\beta = \frac{\pi}{4}$, with the highest $SC = 0.39$, shows a high coherence level with $AMI = 0.9$ even with very few samples (i.e., $N = 100$) (Fig. 4c). This behavior suggests that, for those quantum embeddings that achieve higher scores, the captured structure can be learned with fewer data points.

4.2 Execution on QPU

Moving forward from the ideal simulation, we computed all QKs on a real noisy NISQ device, to assess the reliability of numerical simulation in light of quantum noise and verify how clusters might vary between different embeddings. The current limitations on available systems hinder the ability to run the experiments on the whole dataset; thus, we consider, in the following, a reduced sampling of the data with 100 points. This fact, while it is expected to affect the clustering analysis results, does not undermine the capacity of addressing the overall effects of the quantum noise on the computational pipeline, as it depends mostly on the hardware specification and on the circuit composition. Indeed, to run a quantum circuit on a real quantum computer, specifically a gate base quantum hardware, it is necessary to rewrite it from a high level circuit into an equivalent circuit that matches the topology of the specific quantum device and further to optimize the execution for present day noisy quantum systems. This process, called transpilation, can change substantially the outline of the original circuit in terms of circuit depth, number, and type of gates used. Since these aspects are crucial for determining the resilience of a circuit to noise, they need to be taken into account in discussing quantum hardware results. Specifically, we implemented our experiments on the IBM quantum system *ibmq_Mumbai*, fitting the quantum circuits to the hardware topology as shown in Fig. 5: out of the 27 qubits in the device, we heuristically selected 4 with higher

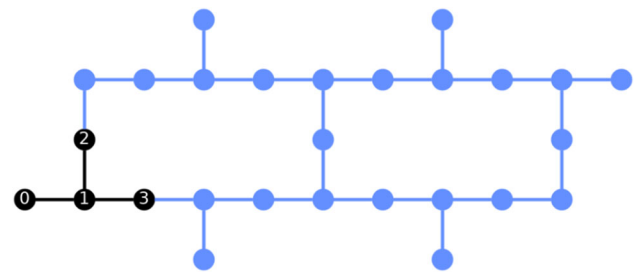


Fig. 5 IBMQ_Mumbai 27 qubits hardware specifics and topology. Qubits selected for circuits implementation are shown in black

level of connectivity. In addition, we used a lightly optimized transpilation, corresponding to a level 1 in the qiskit package, that performs adjacent gate collapsing. In Table 1, we reported the specific of each transpiled circuit in use. Current quantum processors allow for error mitigation techniques; in our experiments, we utilized IBM's error mitigation options by configuring the maximum resilience level available for our setting (i.e., sampler). Specifically, we employed resilience level 1, which mitigates with readout errors.

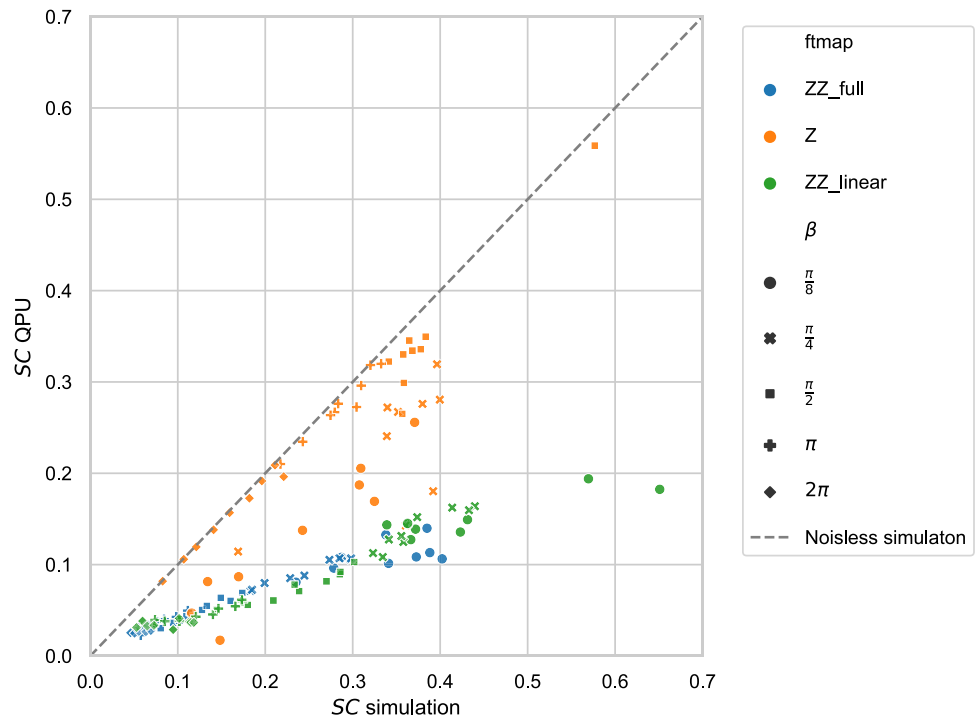
For each QK evaluated on *ibmq_Mumbai*, we calculated the SC, here reported in Fig. 6, versus the values obtained in simulation. The Z feature map with $\beta \geq \frac{\pi}{2}$ exhibits very good consistency with the ideal case, whereas the entangled ZZ linear and ZZ full feature maps deviate more from the ideal scenario. Still, the present behavior can be explained considering the depth and composition of the circuits transpiled on the QPU: in fact, as reported in Table 1, entangled feature maps account for a deeper circuit and a higher number of CNOT gates, thus making them more susceptible to both single qubit noise and gate-level noise. Since different stratification can yield to the same SC values, to further assess the clustering coherence, we measured the AMI between simulated and real quantum stratification for each combination of the feature map, K , and β . For each feature map, we plotted the AMI results for each K and β values in Fig. 7. The deepest feature map (circuit depth = 63), the ZZ linear, presents the lowest AMI scores, while the overall best AMI scores distribution are associated with the Z feature map (circuit depth = 5), which is in principle the less expressive. The fully entangled feature map (circuit depth = 45) shows intermediate values. Thus, the overall trend of the AMI confirms the previous findings in Fig. 6: both the circuits depth and the gate composition emerging from the transpilation pro-

Table 1 This describes the depth and CX gate abundance of the transpiled circuits for each feature map

Feature map (U)	Entanglement type	Circuit depth	CX gate
Z	None	5	0
ZZ	Linear	63	34
ZZ	Full	45	24

The numbers are relative to the full circuit which corresponds to $U^\dagger U$

Fig. 6 Comparison of the SC values obtained on the QPU with the noiseless simulation. SC values from QPU plotted against SC values from simulated runs. Points are colored by feature map and shaped by angle range β . The dotted grey line denotes perfect consistency with the ideal simulation



cess have a significant impact on the clusters consistency. In addition, using the Z feature map the coherence between simulated and real clusters increases with higher values of

β : in particular, for $\beta > \frac{\pi}{4}$ and $K \leq 7$, it achieves perfect agreement or almost perfect agreement. In contrast, the entangled feature maps exhibit the opposite trend, where a

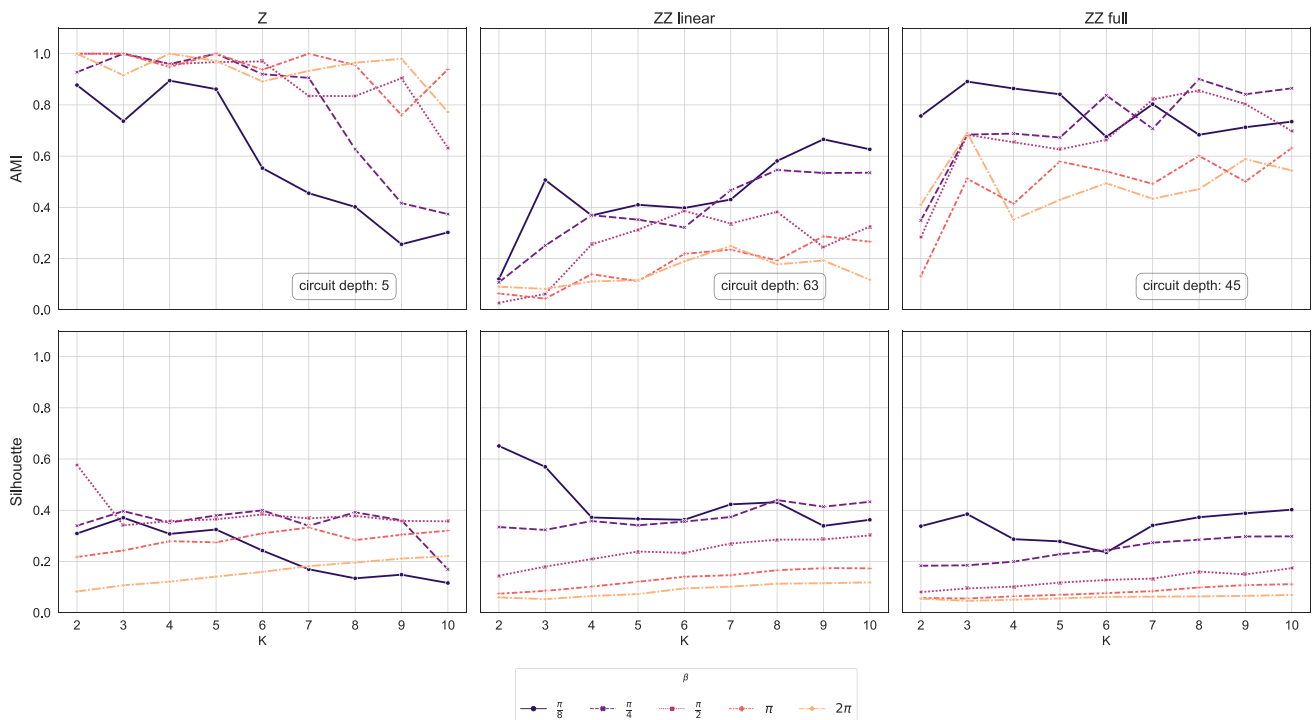


Fig. 7 Plots of AMI comparing simulated and real quantum stratification and SC of ideal simulation for 100 samples. The plots in the figure are organized in rows by metric (top row AMI and bottom SC) and in

columns by feature map type, from left to right: Z, ZZ linear, and ZZ full. The data points are plotted against the number of clusters K and are colored and shaped by angle range β

smaller β corresponds to a higher AMI. By crossing these results from real quantum hardware with the SC obtained in the ideal scenario (Fig. 7), which provides the best possible outcome for these data, it is possible to draw some conclusion about the effect of quantum noise on the clustering. It can be noticed that QKs of entangled feature maps with high β give small SCs; thus, they do not provide accurate clusters. In these cases, the data can be easily partitioned very differently upon perturbations, potentially coming from any noisy source. Hence, when a real quantum hardware is in use, as there is an intrinsic noise source in the computation, the robustness of the clustering provided by the entangled kernel functions, with corresponding low SC values, is strongly hindered by the quantum noise.

4.3 Limitations

Our results show that QKs achieve clustering performance comparable to the classical RBF kernel and a significant reduction in sample complexity; still, these findings should be considered alongside certain limitations of the designed experimental set-up. Specifically, the outcomes are influenced by the data preprocessing step used to reduce dimensionality, but its impact has not been investigated. Future studies should address this in terms of both reduction method and dimension of the projected space. In addition, we acknowledge that, while METABRIC has been chosen as a benchmark for its multi-omics features, high dimensionality, and heterogeneity, our results are dataset dependent and might not uphold to a different multi-omics tumor dataset. In particular, the relationship between the dataset characteristics and the observed advantages of the quantum model, like the reduction in sample complexity, was not quantitatively investigated as it goes beyond the scope of our work, but has emerged as an interesting aspect to study in the future in light of the practical application of QKs.

5 Conclusion and future work

We have investigated QK models for BC subtyping with a real multi-omics dataset addressing the challenges of problem-agnostic quantum embeddings for the best expressivity task-related trade-off. We explored three IPQ-style encoding schemes with different levels of entanglement of the qubits, comparing them with a classical RBF model. The implemented QKs yield a $g_{C|Q}$ that consistently exceeds the threshold, thus guarantying a sufficient separation between the quantum and the classical models. The QK obtained with a noiseless simulation on the full dataset ($N = 1980$) show that the ZZ linear feature map achieved better clustering results, in terms of SC, compared to the classical RBF kernel for finer stratification of BC samples, with $K \geq 9$. In

addition, varying the entanglement level and angle range, we noted how entangled feature map, which are in principle more expressive, strongly benefits from angle tuning, confirming that embedding optimization is needed. Further, we focused on the best data-fitting feature map, the ZZ linear, and demonstrated experimentally that configurations with better clustering results ($SC \geq 0.3$) foster a reduction of sample complexity. Indeed, for the best-performing cases ($K = 2, K = 10$), the ZZ linear was able to obtain highly coherent stratification ($AMI \geq 0.9$) with the one obtained for the full dataset using only 100 samples. This finding, already identified by Li et al. (2019) in the supervised setting, is potentially highly valuable for clinical applications, where datasets and sample cohorts are often not sufficiently large and representative due to experimental and availability limitations. While the proposed methodology showed promising results in crucial aspects for possible future applications of quantum models in biomedicine, we have tested here only a limited subset of possible encoding configurations. To further explore the encoding space to enhance performances, our work could be extended by using a trainable kernel approach (Glick et al. 2021). In such case, a parametrized QK is iteratively adapted to a dataset. In addition, we acknowledged that our results are influenced by the preprocessing choices made for our data. Future work could investigate the impact of dimensionality reduction on performance, exploring both different reduction strategies and varying the dimensionality of the projected space (e.g., extending beyond the 4-dimensional space used in this study). Finally, the execution on a NISQ system, the *ibmq_Mumbai*, showed that quantum noise has little to no effect on less expressive configurations, like the Z feature map that entails shallower circuits, while it hinders the results of entangled feature maps. This behavior suggests that a trade-off between ideal performances and hardware resilience should be considered in QK selection. Nevertheless, it is important to note that NISQ devices now available are rapidly evolving and have reached higher accuracy level, compared to the QPU used in this work (Kim et al. 2023). In addition, a major effort has been made also in advancing noise mitigation strategies (Cai et al. 2022); thus, we expect that, in the foreseeable future, quantum protocols, like the one set out in this work, will achieve sufficient reliability and scalability to be implemented for real applications.

The clustering of tumor samples using multi-omics data is crucial to identify clinically relevant subgroups; still, the reconstruction of non-linear relationship between tumor molecular descriptors is made hard by classical model expressive capacity and limited data. Although QML and QK could offer an alternative view, we faced some major challenges in building proper quantum representation for biomedical systems, where formulating a problem-inspired embedding is not possible, as in Larocca et al. (2022); Liu

et al. (2020); Meyer et al. (2023). Nevertheless, similar considerations apply to classical ML, and given the early stage of quantum approaches, the heuristic exploration of classically complex problems is necessary to identify tasks that could be tackled with current NISQ devices. In this spirit, we would like to further explore the use of QKs for challenging biomedical problems that could benefit from a quantum modeling of the system, such as peptide binding prediction, for which an embedding could be more easily formulated based on a priori knowledge of the system.

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Data availability METABRIC dataset, including gene expression and copy number data, for the 1980 patients is freely available at cBioPortal repository, at the following link: www.cbioportal.org/study/summary?id=brca_metabric.

The data, code, and experimental results associated with this project is available on GitHub at the following link: <https://github.com/ctglab/QKomics>.

Materials availability Not applicable

Declarations

Ethics approval and consent to participate Not applicable

Conflict of interest The authors declare no competing interests.

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References

- Ahalya RK, Snehalatha U, Dhanraj V (2022) Automated segmentation and classification of hand thermal images in rheumatoid arthritis using machine learning algorithms: a comparison with quantum machine learning technique. *J Therm Biol* 111:103404. <https://doi.org/10.2139/ssrn.4191391>
- Aleksandrowicz G, Alexander T, Barkoutsos P et al (2019) Qiskit: an open-source framework for quantum computing. <https://doi.org/10.5281/zenodo.2562111>
- Amin J, Sharif M, Fernandes SL et al (2022) Breast microscopic cancer segmentation and classification using unique 4-qubit-quantum model. *Microsc Res Tech* 85:1926–1936. <https://doi.org/10.1002/jemt.24054>
- Aswiga RV, Sridevi S, Indira B (2024) Leveraging quantum kernel support vector machine for breast cancer diagnosis from digital breast tomosynthesis images. *Quantum Mach Intell*. <https://doi.org/10.1007/s42484-024-00170-3>
- Azevedo V, Silva C, de Castro Dutra I (2022) Quantum transfer learning for breast cancer detection. *Quantum Mach Intell* 4. <https://doi.org/10.1007/s42484-022-00062-4>
- Bernstein ES, Vazirani UV (1993) Quantum complexity theory. *Proceedings of the twenty-fifth annual ACM symposium on theory of computing*. <https://doi.org/10.1145/167088.167097>
- Bersanelli M, Mosca E, Remondini D et al (2016) Methods for the integration of multi-omics data: mathematical aspects. *BMC Bioinforma* 17. <https://doi.org/10.1186/s12859-015-0857-9>
- Biamonte JD, Wittek P, Pancotti N et al (2016) Quantum machine learning. *Nature* 549:195–202. <https://doi.org/10.1038/nature23474>
- Cai Z, Babbush R, Benjamin SC et al (2022) Quantum error mitigation. *Rev Mod Phys*. <https://doi.org/10.1103/RevModPhys.95.045005>
- Canatar A, Peters E, Pehlevan C et al (2022) Bandwidth enables generalization in quantum kernel models. *Trans Mach Learn Res* 2023. <https://doi.org/10.48550/arXiv.2206.06686>
- Caswell-Jin JL, Lorenz C, Curtis C (2021) Molecular heterogeneity and evolution in breast cancer. <https://www.annualreviews.org/content/journals/10.1146/annurev-cancerbio-060220-014137>
- Cerezo M, Verdon G, Huang HY et al (2022) Challenges and opportunities in quantum machine learning. *Nat Comput Sci* 2:567–576. <https://doi.org/10.1038/s43588-022-00311-3>
- Cordier BA, Sawaya NPD, Guerreschi GG et al (2021) Biology and medicine in the landscape of quantum advantages. *J R Soc Interface* 19. <https://doi.org/10.1098/rsif.2022.0541>
- Curtis C, Shah SP, Chin SF et al (2012) The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups. *Nature* 486:346–352. <https://doi.org/10.1038/nature10983>
- Di Marcantonio F, Incudini M, Tezza D et al (2023) Quantum advantage seeker with kernels (quask): a software framework to accelerate research in quantum machine learning. *Quantum Mach Intell* 5(1):20. <https://link.springer.com/article/10.1007/s42484-023-00107-2>
- Dunjko V, Briegel HJ (2018) Machine learning & artificial intelligence in the quantum domain: a review of recent progress. *Rep Prog Phys* 81. <https://doi.org/10.1088/1361-6633/aab406>
- Flother FF (2023) The state of quantum computing applications in health and medicine

- Gil-Fuster E, Eisert J, Dunjko V (2023) On the expressivity of embedding quantum kernels. *Mach Learn Sci Technol* 5. <https://doi.org/10.1088/2632-2153/ad2f51>
- Glick JR, Gujarati TP, Córcoles AD et al (2021) Covariant quantum kernels for data with group structure. *Nat Phys* 20:479–483. <https://doi.org/10.1038/s41567-023-02340-9>
- Havlicek V, Córcoles AD, Temme K et al (2018) Supervised learning with quantum-enhanced feature spaces. *Nature* 567:209–212. <https://doi.org/10.1038/s41586-019-0980-2>
- Huang HY, Broughton M, Mohseni M et al (2020) Power of data in quantum machine learning. *Nat Commun* 12. <https://doi.org/10.1038/s41467-021-22539-9>
- Includini M, Martini F, Pierro AD (2024) Toward useful quantum kernels. *Adv Quantum Technol*. <https://doi.org/10.1002/qute.202300298>
- Includini M, Bosco DL, Martini FD et al (2022). Automatic and effective discovery of quantum kernels. <https://doi.org/10.1109/TETCI.2024.3499993>
- Kim Y, Eddins A, Anand S et al (2023) Evidence for the utility of quantum computing before fault tolerance. *Nature* 618:500–505. <https://doi.org/10.1038/s41586-023-06096-3>
- Krunic ZO, Flöther FF, Seegan GW et al (2021) Quantum kernels for real-world predictions based on electronic health records. *IEEE Trans Quantum Eng* 3:1–11. <https://doi.org/10.1109/TQE.2022.3176806>
- Larocca M, de Sauvage FJ, Sbahi FM et al (2022) Group-invariant quantum machine learning. *PRX Quantum* 3:030341. <https://doi.org/10.1103/PRXQuantum.3.030341>. <https://link.aps.org/doi/10.1103/PRXQuantum.3.030341>
- Li RY, Gujja S, Bajaj SR et al (2019) Quantum processor-inspired machine learning in the biomedical sciences. *Patterns* 2. <https://doi.org/10.1016/j.patter.2021.100246>
- Liu Y, Arunachalam S, Temme K (2020) A rigorous and robust quantum speed-up in supervised machine learning. *Nat Phys* 17:1013–1017. <https://doi.org/10.1038/s41567-021-01287-z>
- Liu Y, Li Z, Xiong H et al (2010) Understanding of internal clustering validation measures. 2010 IEEE international conference on data mining, pp 911–916. <https://doi.org/10.1109/ICDM.2010.35>
- McInnes L, Healy J, Saul N et al (2018) Umap: uniform manifold approximation and projection. *J Open Source Softw* 3(29):861. <https://doi.org/10.21105/joss.00861>
- Mengoni R, Pierro AD (2019) Kernel methods in quantum machine learning. *Quantum Mach Intell* 1:65–71. <https://doi.org/10.1007/s42484-019-00007-4>
- Meyer JJ, Mularski M, Gil-Fuster E et al (2023) Exploiting symmetry in variational quantum machine learning. *PRX Quantum* 4:010328. <https://doi.org/10.1103/PRXQuantum.4.010328>. <https://link.aps.org/doi/10.1103/PRXQuantum.4.010328>
- Moradi S, Brandner C, Spielvogel CP et al (2021). Clinical data classification with noisy intermediate scale quantum computers. <https://doi.org/10.1038/s41598-022-05971-9>
- Nguyen XV, Epps J, Bailey J (2009) Information theoretic measures for clusterings comparison: is a correction for chance necessary? In: International conference on machine learning. <https://doi.org/10.1145/1553374.1553511>
- Park JE, Quanz B, Wood SP et al (2020) Practical application improvement to quantum svm: theory to practice. [arXiv:2012.07725](https://arxiv.org/abs/2012.07725)
- Pedregosa F, Varoquaux G, Gramfort A et al (2011) Scikit-learn: machine learning in Python. *J Mach Learn Res* 12:2825–2830
- Perdomo-Ortiz A, Benedetti M, Realpe-Gómez J et al (2017) Opportunities and challenges for quantum-assisted machine learning in near-term quantum computers. *Quantum Sci Technol* 3. <https://doi.org/10.1088/2058-9565/aab859>
- Ronan T, Qi Z, Naegle KM (2016) Avoiding common pitfalls when clustering biological data. *Sci Signal* 9:re6–re6. <https://doi.org/10.1126/scisignal.aad1932>
- Rousseeuw PJ (1987) Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *J Comput Appl Math* 20:53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- Russnes HG, Lingjærde OC, Børresen-Dale AL et al (2017) Breast cancer molecular stratification: From intrinsic subtypes to integrative clusters. *Am J Pathol* 187(10):2152–2162. <https://doi.org/10.1016/j.ajpath.2017.04.022>
- Scholkopf B, Smola A (2001) Learning with Kernels: support vector machines, regularization, optimization, and beyond. In: Adaptive computation and machine learning series. <https://doi.org/10.7551/mitpress/4175.001.0001>
- Schuld M (2021) Quantum machine learning models are kernel methods. https://doi.org/10.1007/978-3-030-83098-4_6
- Schuld M, Killoran N (2018) Quantum machine learning in feature Hilbert spaces. *Phys Rev Lett* 122(4):040504. <https://doi.org/10.1103/PhysRevLett.122.040504>
- Shaydulin R, Wild SM (2022) Importance of kernel bandwidth in quantum machine learning. *Phys Rev A* 106:042407. <https://doi.org/10.1103/PhysRevA.106.042407>. <https://link.aps.org/doi/10.1103/PhysRevA.106.042407>
- Shor PW (1994) Algorithms for quantum computation: discrete logarithms and factoring. Proceedings 35th annual symposium on foundations of computer science, pp 124–13. <https://doi.org/10.1109/SFCS.1994.365700>
- Siegel RL, Miller KD, Jemal A (2020) Cancer statistics, 2020. *CA Cancer J Clin* 70. <https://doi.org/10.3322/caac.21590>
- Slattery L, Shaydulin R, Chakrabarti S et al (2023) Numerical evidence against advantage with quantum fidelity Kernels on classical data. *Phys Rev A* 107:062417. <https://doi.org/10.1103/PhysRevA.107.062417>. <https://link.aps.org/doi/10.1103/PhysRevA.107.062417>
- Suzuki Y, Yano H, Gao Q et al (2019) Analysis and synthesis of feature map for Kernel-based quantum classifier. *Quantum Mach Intell* 2:1–9. <https://doi.org/10.1007/s42484-020-00020-y>
- Thanasilp S, Wang S, Cerezo M et al (2022) Exponential concentration in quantum Kernel methods. *Nat Commun* 15. <https://doi.org/10.1038/s41467-024-49287-w>
- Tran KA, Kondrashova O, Bradley A et al (2021) Deep learning in cancer diagnosis, prognosis and treatment selection. *Genome Med* 13. <https://doi.org/10.1186/s13073-021-00968-x>
- Vashisth S, Dhall I, Aggarwal G (2021) Design and analysis of quantum powered support vector machines for malignant breast cancer diagnosis. *J Intell Syst* 30:998–1013. <https://doi.org/10.1515/jisys-2020-0089>
- Weinstein JN, Collis EA, Mills GB et al (2013) The cancer genome atlas pan-cancer analysis project. *Nat Genet* 45:1113–1120. <https://www.nature.com/articles/ng.2764#article-info>
- Williams CKI (2001) Learning with Kernels: support vector machines, regularization, optimization, and beyond. *J Am Stat Assoc* 98:489–489. <https://doi.org/10.1198/jasa.2003.s269>

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